

Section D. Model specifications and diagnostics

Setup

```
packages <- c("readxl", "forecast", "dlnm", "splines", "Metrics",
             "demography", "tseries", "seastests")
invisible(lapply(packages, library, character.only = TRUE))

find_project_root <- function(start = getwd()) {
  path <- normalizePath(start, mustWork = FALSE)
  for (i in seq_len(8)) {
    if (dir.exists(file.path(path, "Code")) && dir.exists(file.path(path, "Data"))) {
      return(path)
    }
    parent <- dirname(path)
    if (identical(parent, path)) break
    path <- parent
  }
  stop("Set LC_DLNM_BASE_DIR to the project root.")
}

base_dir <- Sys.getenv("LC_DLNM_BASE_DIR")
if (!nzchar(base_dir)) base_dir <- find_project_root()

function_dir <- file.path(base_dir, "Code/Function")
results_dir <- file.path(base_dir, "Results/Diagnostics")
calibration_rmd <- file.path(
  base_dir,
  "Code/Main_Paper/Section_4_Empirical_results/01_DLNM_LC_LL_calibration.Rmd"
)
calibration_results_dir <- file.path(base_dir, "Results/Calibration")
dir.create(results_dir, recursive = TRUE, showWarnings = FALSE)
dir.create(calibration_results_dir, recursive = TRUE, showWarnings = FALSE)
files <- list.files(function_dir, pattern = "\\..R$", full.names = TRUE)
invisible(sapply(files, source))
```

D.1 Optimal time series model for time-varying factors

```
ensure_sarima_tables <- function() {
  required_files <- file.path(
    calibration_results_dir,
    c("optimal_sarima_table.tex", "optimal_sarima_coef_table.tex")
  )

  if (all(file.exists(required_files))) {
    return(invisible(required_files))
  }
}
```

```

if (!file.exists(calibration_rmd)) {
  stop("Calibration Rmd not found: ", calibration_rmd)
}

old_unnamed_label <- knitr::opts_knit$get("unnamed.chunk.label")
knitr::opts_knit$set(unnamed.chunk.label = "calibration_component")
on.exit(knitr::opts_knit$set(unnamed.chunk.label = old_unnamed_label),
  add = TRUE)

script <- tempfile(fileext = ".R")
knitr::purl(calibration_rmd, output = script, quiet = TRUE)
calibration_env <- new.env(parent = globalenv())
invisible(capture.output(sys.source(script, envir = calibration_env)))

if (!all(file.exists(required_files))) {
  build_sarima_tables_from_fits(
    calibration_env$b,
    calibration_env$b1,
    calibration_env$b2,
    calibration_env$b3,
    calibration_results_dir
  )
}

missing_files <- required_files[!file.exists(required_files)]
if (length(missing_files) > 0) {
  stop("Could not generate SARIMA table file(s): ",
    paste(basename(missing_files), collapse = ", "))
}

invisible(required_files)
}

sarima_files <- ensure_sarima_tables()
cat(readLines(sarima_files[[1]]), sep = "\n")

```

Time series model	DLNM-LC	DLNM-LL
$K(t)$	–	$\text{ARIMA}(1, 0, 2) \times (0, 0, 1)_{52}$
$\kappa(t, \text{Athens})$	$\text{ARIMA}(1, 0, 0) \times (0, 0, 0)_{52}$	$\text{ARIMA}(2, 0, 2) \times (0, 0, 1)_{52}$
$\kappa(t, \text{Lisbon})$	$\text{ARIMA}(1, 0, 0) \times (1, 0, 0)_{52}$	$\text{ARIMA}(1, 0, 0) \times (0, 0, 0)_{52}$
$\kappa(t, \text{Rome})$	$\text{ARIMA}(2, 0, 1) \times (1, 0, 0)_{52}$	$\text{ARIMA}(2, 0, 2) \times (1, 0, 0)_{52}$

Table 1: The selected optimal SARIMA model for time-varying factors.

```
cat("\n\n")
```

```
cat(readLines(sarima_files[[2]]), sep = "\n")
```

DLNM-LC	$\hat{\phi}_1$	$\hat{\phi}_2$	$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\Phi}_1$	$\hat{\Theta}_1$
$\kappa(t, \text{Athens})$	0.282***	—	—	—	—	—
$\kappa(t, \text{Lisbon})$	0.487***	—	—	—	0.150**	—
$\kappa(t, \text{Rome})$	1.067***	-0.103	-0.842***	—	0.102	—
DLNM-LL	$\hat{\phi}_1$	$\hat{\phi}_2$	$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\Phi}_1$	$\hat{\Theta}_1$
$K(t)$	0.762***	—	-0.320**	0.008	—	0.108*
$\kappa(t, \text{Athens})$	0.414	0.247	-0.055	-0.299*	—	0.145**
$\kappa(t, \text{Lisbon})$	0.339***	—	—	—	—	—
$\kappa(t, \text{Rome})$	1.539***	-0.560	-1.389***	0.435	-0.066	—

Level of significance: 0.01:***, 0.05:**, 0.1:*

Table 2: The coefficients of selected optimal SARIMA models on time-varying factors.

D.2 Seasonality test on residuals

Run Proposed Models

```
diagnostics <- DLNM_residual_diagnostics(base_dir)
```

Helper Functions

```
format_p <- function(x) {
  ifelse(is.na(x), "--", sprintf("%.2f", as.numeric(x)))
}

make_seasonality_latex <- function(test_list, display_names, model_name,
                                   caption, label) {
  region_order <- c("Attiki", "Lisbon", "Roma")
  region_order <- region_order[region_order %in% names(test_list)]
  test_rows <- c(QS = "QS",
                 Friedman = "Friedman",
                 `K--W` = "Kruskal-Wallis")

  lines <- c(
    "\\begin{table}[H]",
    "\\centering",
    "\\begin{tabular}{lccccccc}",
    "\\toprule",
    paste0(
      "\\multirow{2}{*}{\\textbf{"}, model_name, "}"",
      " & \\multicolumn{3}{c}{\\textbf{"}, display_names[[region_order[[1]]]], "}"",
      " & \\multicolumn{3}{c}{\\textbf{"}, display_names[[region_order[[2]]]], "}"",
      " & \\multicolumn{3}{c}{\\textbf{"}, display_names[[region_order[[3]]]], "}"",
    ),
    "\\cmidrule(lr){2-4} \\cmidrule(lr){5-7} \\cmidrule(lr){8-10}",
    paste0(
      " & \\textbf{QS} & \\textbf{Friedman} & \\textbf{K--W}",
      " & \\textbf{QS} & \\textbf{Friedman} & \\textbf{K--W}",
      " & \\textbf{QS} & \\textbf{Friedman} & \\textbf{K--W} \\",
    ),
    "\\midrule"
```

```

)

for (age_group in colnames(test_list[[region_order[[1]]]])) {
  values <- c()
  for (region in region_order) {
    test_table <- test_list[[region]]
    values <- c(values, format_p(test_table[test_rows, age_group]))
  }
  age_label <- gsub("-", "--", age_group)
  lines <- c(lines, paste0(age_label, " & ", paste(values, collapse = " & "), " \\\\"))
}

lines <- c(
  lines,
  "\\bottomrule",
  "\\end{tabular}",
  paste0("\\caption{", caption, "}"),
  paste0("\\label{", label, "}"),
  "\\end{table}"
)
paste(lines, collapse = "\\n")
}

```

DLNM-LC Residual Seasonality

```

lc_tests <- diagnostics$DLNM_LC$seasonality_tests
lc_display_names <- diagnostics$DLNM_LC$display_names

lc_latex <- make_seasonality_latex(
  lc_tests,
  lc_display_names,
  "DLNM--LC",
  "The seasonality test of model residual in DLNM--LC model.",
  "tab:seasonality_dlnm_lc"
)

writeLines(lc_latex, file.path(results_dir, "residual_seasonality_DLNM_LC.tex"))
cat(lc_latex)

```

DLNM-LC	Athens			Lisbon			Rome		
	QS	Friedman	K-W	QS	Friedman	K-W	QS	Friedman	K-W
20-64	1.00	0.36	0.34	1.00	0.95	0.81	1.00	0.66	0.20
65-74	1.00	0.08	0.03	0.26	0.53	0.10	0.21	0.03	0.01
75-84	1.00	1.00	0.96	1.00	0.97	0.95	1.00	0.93	0.80
85+	1.00	0.47	0.66	1.00	0.06	0.04	1.00	0.80	0.64

Table 3: The seasonality test of model residual in DLNM-LC model.

DLNM-LL Residual Seasonality

```

ll_tests <- diagnostics$DLNM_LL$seasonality_tests
ll_display_names <- diagnostics$DLNM_LL$display_names

ll_latex <- make_seasonality_latex(
  ll_tests,
  ll_display_names,
  "DLNM--LL",
  "The seasonality test of model residual in DLNM--LL model.",
  "tab:seasonality_dlnm_ll"
)

writeLines(ll_latex, file.path(results_dir, "residual_seasonality_DLNM_LL.tex"))
cat(ll_latex)

```

DLNM-LL	Athens			Lisbon			Rome		
	QS	Friedman	K-W	QS	Friedman	K-W	QS	Friedman	K-W
20-64	0.30	0.28	0.24	1.00	0.91	0.82	1.00	0.58	0.22
65-74	1.00	0.10	0.03	0.27	0.48	0.09	0.29	0.06	0.01
75-84	1.00	1.00	0.98	1.00	0.97	0.96	1.00	0.89	0.73
85+	1.00	0.47	0.56	1.00	0.08	0.04	1.00	0.74	0.51

Table 4: The seasonality test of model residual in DLNM-LL model.